OECD/NEA BENCHMARK BASED ON NUPEC PWR SUBCHANNEL AND BUNDLE TESTS (PSBT)

Assembly Specifications and Benchmark Database

Volume I

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Incorporated Administrative Agency
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(JNES)

CONTENTS

1. Introduction

Void fraction measurements[1-3] and departure from nucleate boiling (DNB) tests[1, 4] were performed under the conditions simulating PWR thermal-hydraulic conditions including the steady states and the transients such as the power increase, the flow reduction, the depressurization and the temperature increase.

2 Void Fraction Measurements

2.1 Test Facility

The void fraction measurement tests for PWR fuel assemblies have been performed since 1987 to 1993. This test contains the subchannel experiments and the rod bundle experiments. The void fraction in each experiment is measured by the gamma-ray transmission method.

Figure 2.1-1 is a system diagram of the test facility[5]. The subchannel test section, as shown in Figure 2.1-2, simulates one of subchannels of a PWR fuel assembly. The effective heated length is 1500 mm where the void measuring section is set near the top end at 1400 mm from the bottom of the heated section.

Figure 2.1-3 shows the rod bundle test section. Test bundles used in the rod bundle test are fabricated simulating partial section and full length of the 17×17 type PWR fuel assembly. These are arranged in 5×5 square array. The effective heated length is 3658 mm where the void measuring sections are set 2216 mm (Lower), 2669 mm (Middle) and 3177 mm (Upper) each from a bottom of the heated length.

The transmission method of gamma-ray is adopted in this study to measure the density and converted to the void fraction of the gas-liquid two-phase flow. Figure 2.1-4 shows the test procedure. The detailed explanation for the void fraction measurement procedure is provided in References (1) and (2).

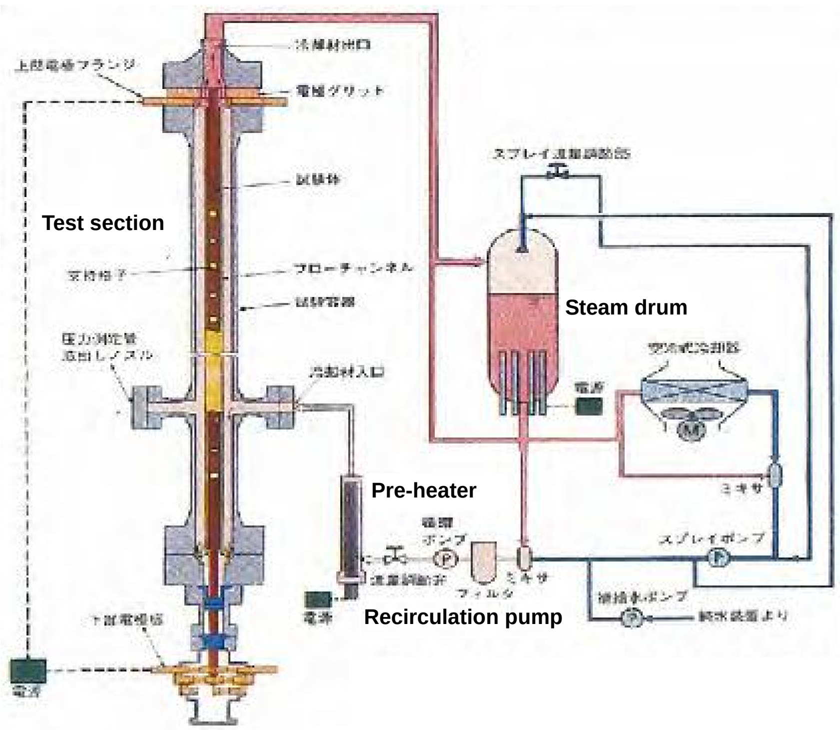


Figure 2.1-1 System diagram of test facility for NUPEC PWR test series

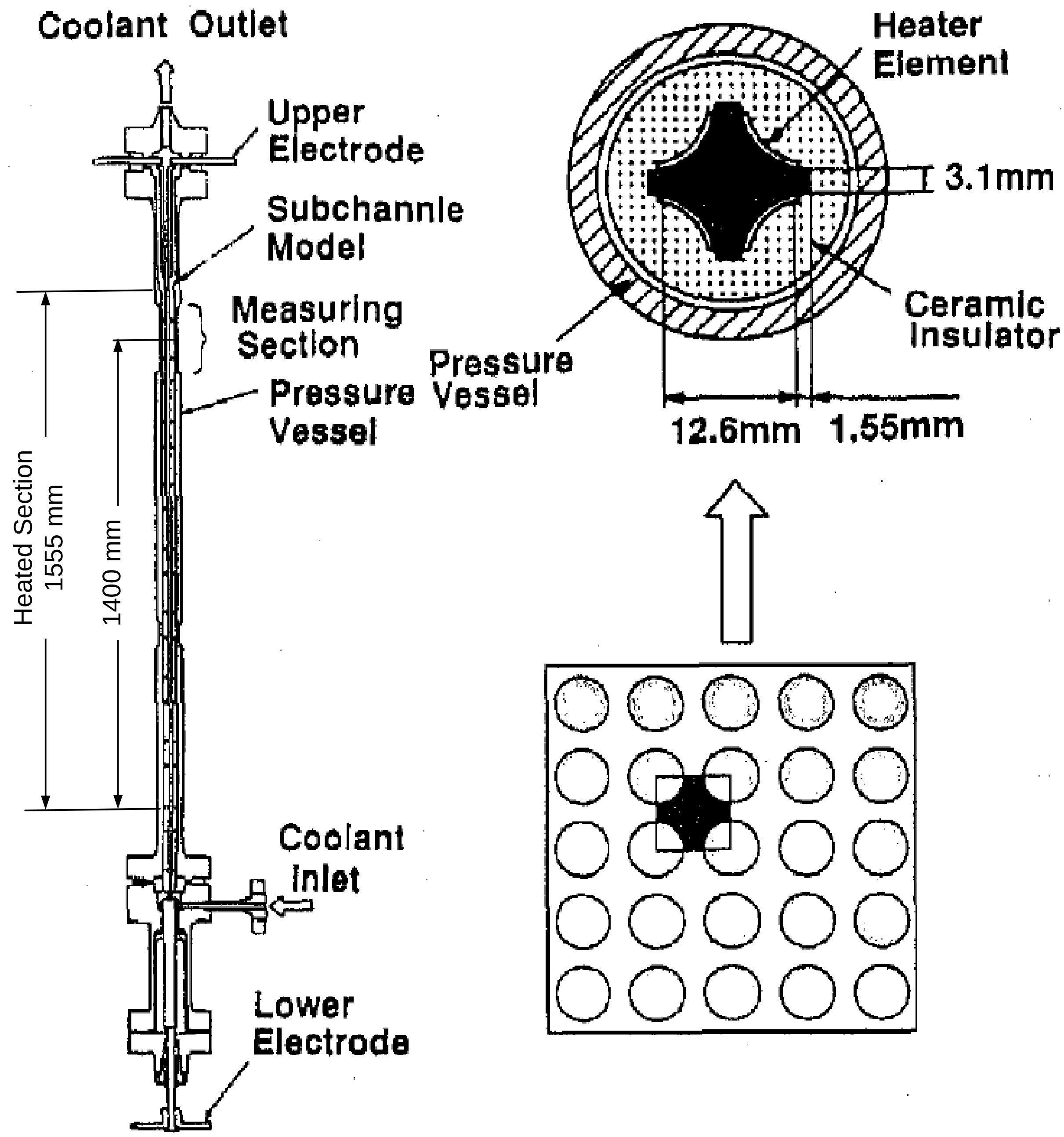


Figure 2.1-2 Test section for subchannel void measurement

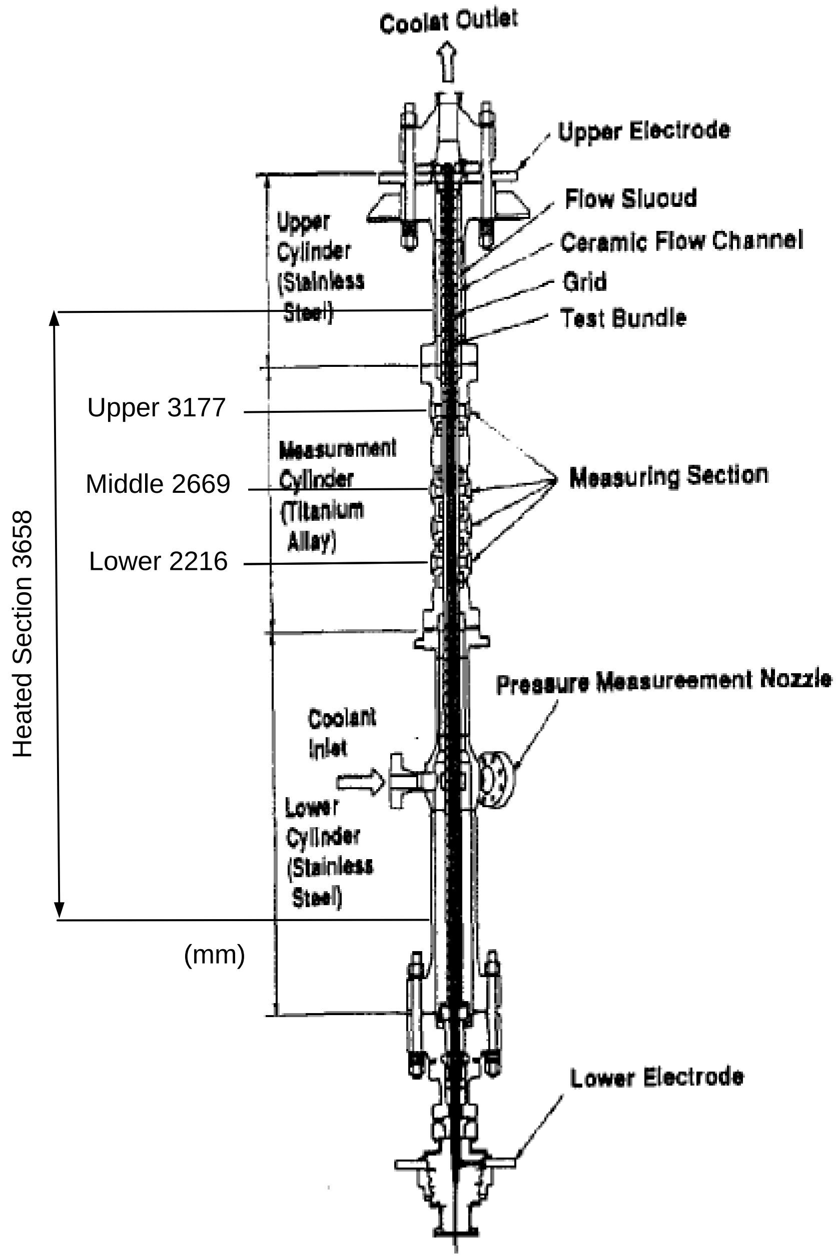
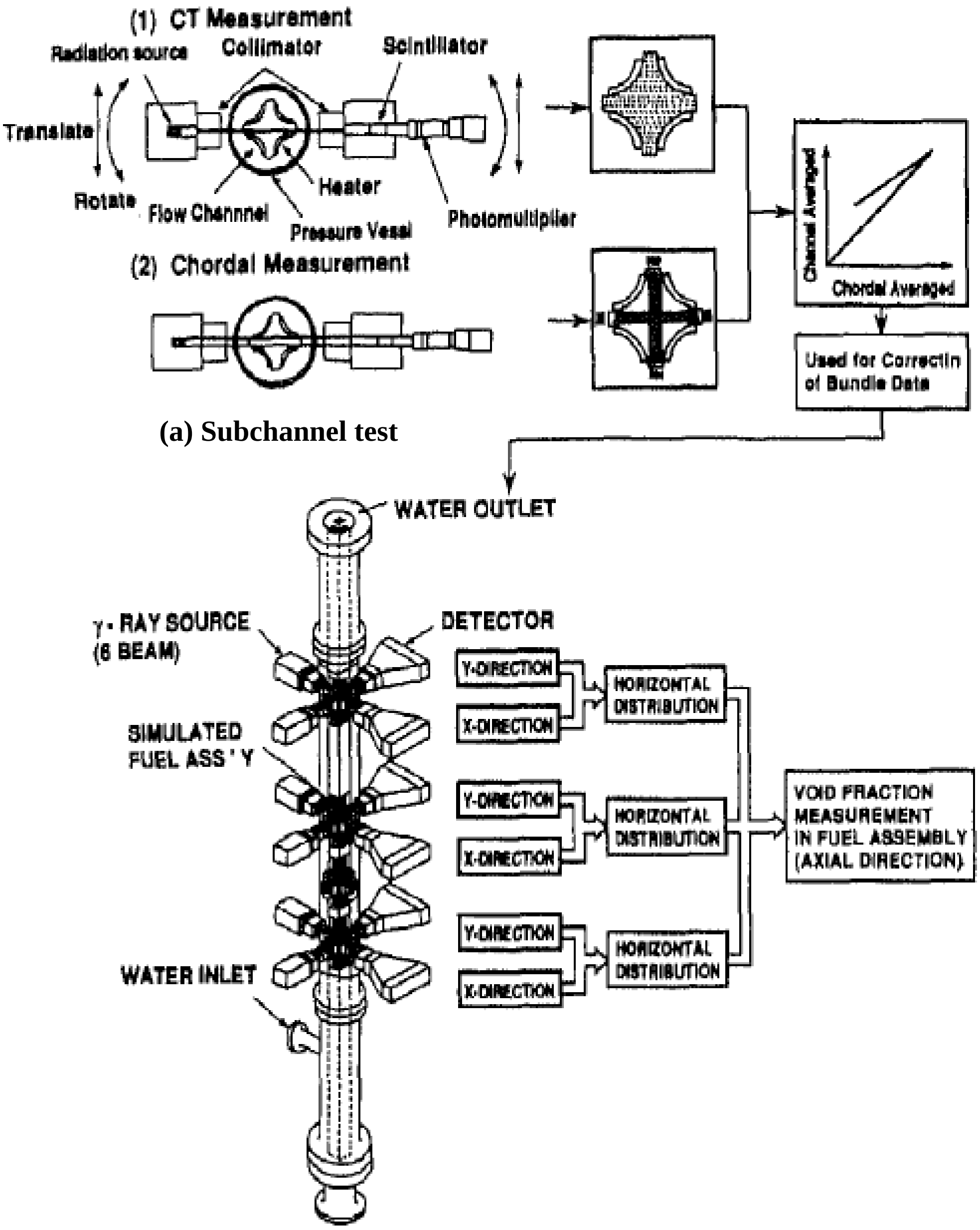


Figure 2.1-3 Test section for rod bundle void measurement



(a) Subchannel test

(b) Rod bundlel test

Figure 2.1-4 Void fraction measurement procedure

Table 2.1-1 summarizes the estimated accuracies of the major process parameters in this test series.

Calibration data were measured in the following conditions for both CT and Chordal measurements.

- Assembly filled with air under room temperature atmospheric condition
- Assembly filled with water under room temperature atmospheric condition

after the heating and pressurize up the test loop,

- Assembly filled with water under high temperature high pressure condition

The gamma-ray source was ^{137}Cs and the intensity of it was designed so that the measured count-rate of the signal is 3×10^5 cps.

Table 2.1-1 Estimated accuracy for void fraction measurements

Quantity	Accuracy
Process parameters	
Pressure	1 %
Flow	1.5%
Power	1 %
Fluid temperature	1 Celsius
Void fraction measurement	
CT measurement	
Gamma-ray beam width	1 mm
Subchannel averaged (steady state)	3% void
Spatial resolution of one pixel	0.5 mm
Chordal measurement	
Gamma-ray beam width (center)	3 mm
Gamma-ray beam width (side)	2 mm
Subchannel averaged (steady state)	4% void
Subchannel averaged (transient)	5% void

Table 2.1-2 Number of Gamma-ray beam

Test assembly	CT measurement	Chordal measurement
Subchannel	2 (X and Y direction)	2 (X and Y direction)
Rod bundle	-	6 beam × 2 × 3 section (total 36 beams)

Table 2.1-3 Time needed for void fraction measurements

Item		CT measurement	Chordal measurement
Steady-state	Time needed	5 s/step×^{T}33×^{R} 17 step (it takes 2 h)	100 s (sampling cycle 0.1 s)
	Measurement	2 times	3 times
Transient	Time needed	-	200 s
	Measurement	-	1 time

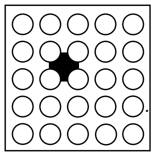
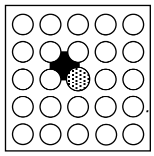
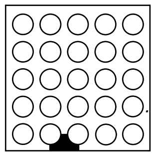
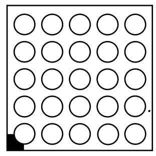
2.2 Test assembly

Table 2.2-1 Test assembly for void fraction measurement

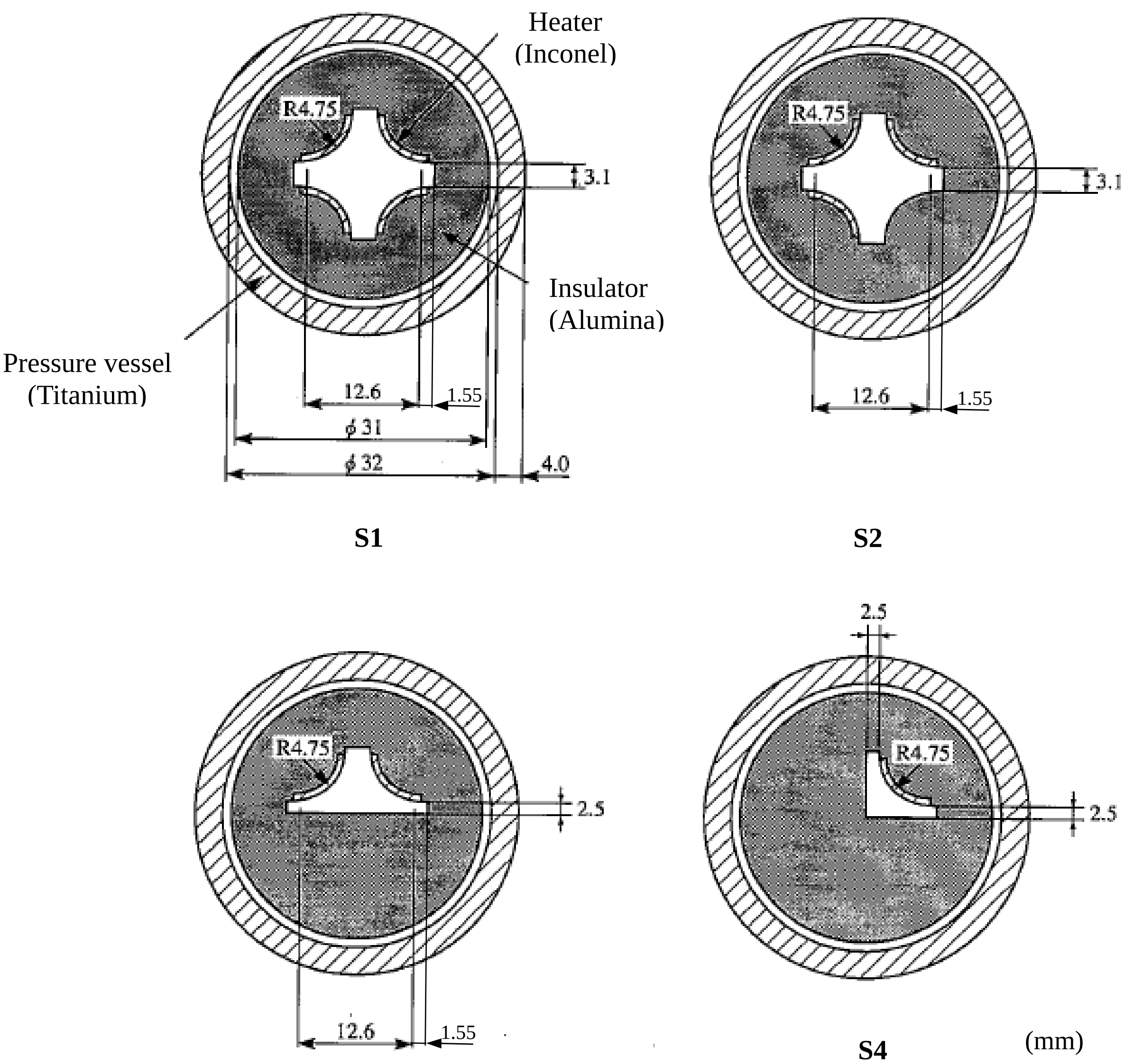
Assembly	Reference fuel type	Type of cell		Power distribution	
				Radial	Axial
S1	17×17 M	Subchannel	Center (Typical)	-	Uniform
S2			Center (Thimble)	-	Uniform
S3			Side	-	Uniform
S4			Corner	-	Uniform
B5	17×17 M	5×5 Rod bundle	Typical cell	A	Uniform
B6			Typical cell	A	Cosine
B7			Thimble cell	B	Cosine

(1) Geometry and power shape

Table 2.2-2 Geometry and power shape of subchannel test assembly

Item	Data			
Assembly (Subjected subchannel)	 **S1**	 **S2**	 **S3**	 **S4**
Subchannel type	Center (Typical)	Center (Thimble)	Side	Corner
Number of heaters	4×1/4	3×1/4	2×1/4	1×1/4
Axial heated length (mm)	1555	1555	1555	1555
Axial power shape	Uniform	Uniform	Uniform	Uniform

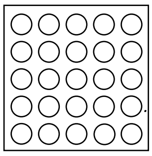
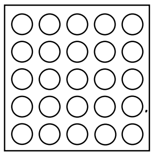
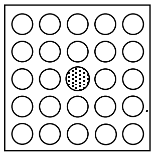
■: Subjected subchannel ○ : Heated rod ◍ : Thimble rod



S3

Figure 2.2-1 Cross-sectional view of subchannel test assembly

Table 2.2-3 Geometry and power shape of rod bundle test assembly

Item	Data		
Assembly	 B5	 B6	 B7
Rods array	5×5	5×5	5×5
Number of heated rods	25	25	24
Number of thimble rods	0	0	1
Heated rod outer diameter (mm)	9.50	9.50	9.50
Thimble rod outer diameter (mm)	-	-	12.24
Heated rods pitch (mm)	12.60	12.60	12.60
Axial heated length (mm)	3658	3658	3658
Flow channel inner width (mm)	64.9	64.9	64.9
Radial power shape	A	A	B
Axial power shape	Uniform	Cosine	Cosine
Number of MV spacers	7	7	7
Number of NMV spacers	2	2	2
Number of simple spacers	8	8	8
MV spacer location (mm)	471, 925, 1378, 1832, 2285, 2739, 3247		
NMV spacer location (mm)	2.5, 3755		
Simple spacer location (mm)	237, 698, 1151, 1605, 2059, 2512, 2993, 3501		

○ : Heated rod ◍ : Thimble rod
MV: Mixing vane, NMV: No mixing vane
Spacer location is distance from bottom of heated length to spacer bottom face.

Table 2.2-4 Tolerance of manufacturing for test assembly

Item	Tolerance
Rod bundle	
Heater rod diameter	0.02 mm
Heater rod displacement	0.45 mm
Flow channel inner width	0.05 mm
Flow channel displacement	0.20 mm
Power distribution	3 %

Table 2.2-5 Radial power distribution

Pattern A

0.85	0.85	0.85	0.85	0.85
0.85	1	1	1	0.85
0.85	1	1	1	0.85
0.85	1	1	1	0.85
0.85	0.85	0.85	0.85	0.85

Pattern B

0.85	0.85	0.85	0.85	0.85
0.85	1	1	1	0.85
0.85	1	0	1	0.85
0.85	1	1	1	0.85
0.85	0.85	0.85	0.85	0.85

Table 2.2-6 Axial power shape

Node	Relative power
	Cosine
(Bottom)	
1	0.42
2	0.47
3	0.56
4	0.67
5	0.80
6	0.94
7	1.08
8	1.22
9	1.34
10	1.44
11	1.51
12	1.55
13	1.55
14	1.51
15	1.44
16	1.34
17	1.22
18	1.08
19	0.94
20	0.80
21	0.67
22	0.56
23	0.47
24	0.42
(Top)	

(2) Heater rod structure

Heater rod structure for subchannel test assembly is specified in Table 2.2-7.

Table 2.2-7 Heater rod structure for subchannel test assembly

Item		Data
Heater	Outer radius (mm)	4.75
	Thickness (mm)	0.85
	Material	Inconel 600
	Heating method	Direct heating
Insulator	Outer diameter (mm)	31
	Material	Alumina
Pressure vessel	Inner diameter (mm)	32
	Thickness (mm)	4.0
	Material	Titanium

Heater rod structure for rod bundle test assembly is specified in Table 2.2-8.

Table 2.2-8 Heater rod structure for bundle test assembly

Item		Data
Heater	Outer diameter (mm)	9.5
	Thickness (mm)	0.65
	Material	Inconel 600
	Heating method	Direct heating
Insulator	Outer diameter (mm)	8.2
	Inner diameter (mm)	5.8
	Material	Alumina

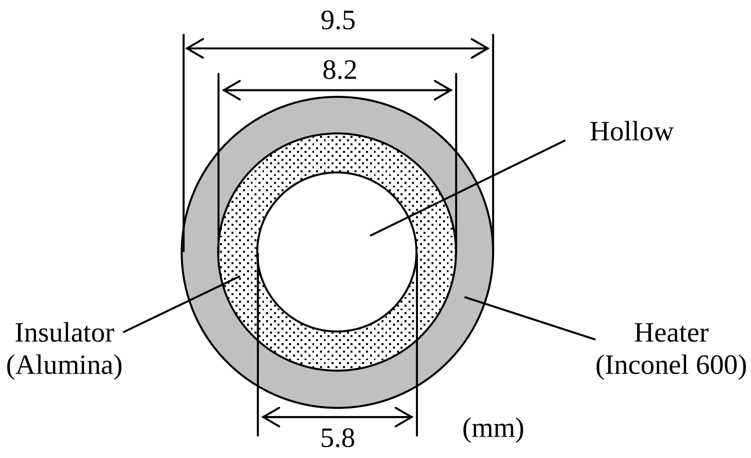


Figure 2.2-2 Cross-sectional view of heater rod for bundle test assembly

(3) Location of pressure taps

The bundle pressure drop was monitored at several locations as depicted in Figure 2.2-3.

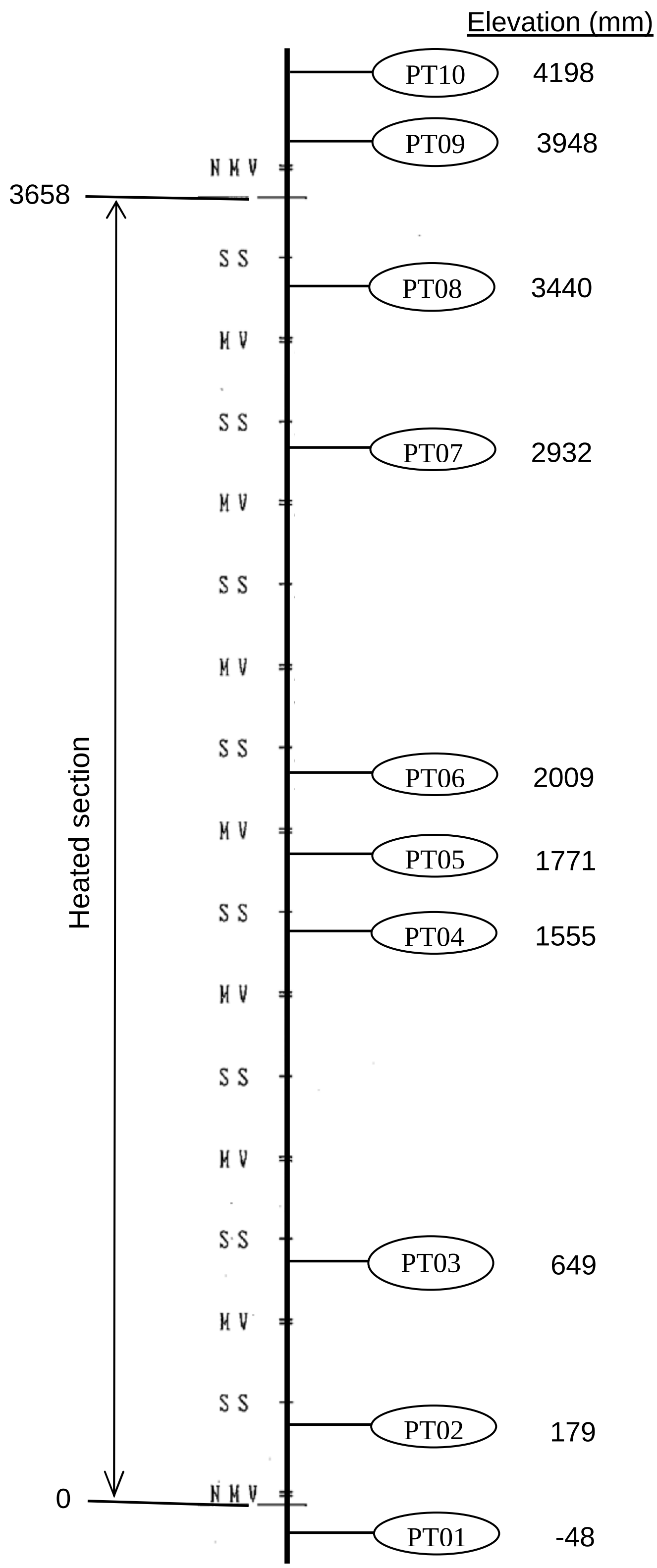


Figure 2.2-3 Location of pressure taps

3. DNB Measurements

3.1 Test Facility

The DNB measurement tests for PWR fuel assemblies have been performed since 1983 to 1989. These tests were conducted with full length partial array of rod bundles simulating the 17×17 type PWR fuel assembly under the conditions including the steady states and the transients.

Table 3.1-1 summarizes the estimated accuracies of the major process parameters in this test series.

Table 3.1-1　Estimated accuracy for DNB measurements

Quantity	Accuracy
Process parameters	
Pressure	1 %
Flow	1.5%
Power	1 %
Fluid temperature	1 Celsius

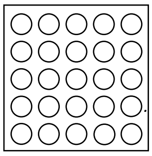
3.2 Test Assembly

Table 3.2-1　Test assembly for DNB measurements

<table>
<tr><th rowspan="2">Assembly</th><th rowspan="2">Reference fuel type</th><th rowspan="2">Rods array</th><th rowspan="2">Type of cell</th><th colspan="2">Power distribution</th></tr>
<tr><th>Radial</th><th>Axial</th></tr>
<tr><td>A0</td><td rowspan="8">17×17 M</td><td rowspan="3">5×5</td><td>Typical cell</td><td>A</td><td>Uniform</td></tr>
<tr><td>A1</td><td>Typical cell</td><td>C</td><td>Uniform</td></tr>
<tr><td>A2</td><td>Typical cell</td><td>A</td><td>Uniform</td></tr>
<tr><td>A3</td><td>6×6</td><td>Typical cell</td><td>D</td><td>Uniform</td></tr>
<tr><td>A4</td><td rowspan="4">5×5</td><td>Typical cell</td><td>A</td><td>Cosine</td></tr>
<tr><td>A8</td><td>Thimble cell</td><td>B</td><td>Cosine</td></tr>
<tr><td>A11</td><td>Typical cell</td><td>A</td><td>Cosine</td></tr>
<tr><td>A12</td><td>Thimble cell</td><td>B</td><td>Cosine</td></tr>
</table>

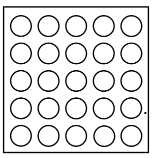
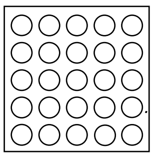
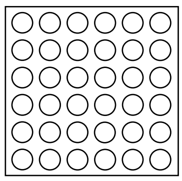
(1) Geometry and power shape

Table 3.2-2 Geometry and power shape of test assembly A0

Item	Data
Assembly	 **A0**
Rods array	5×5
Number of heated rods	25
Number of thimble rods	0
Heated rod outer diameter (mm)	9.50
Thimble rod outer diameter (mm)	-
Heated rods pitch (mm)	12.60
Axial heated length (mm)	3658
Flow channel inner width (mm)	64.9
Radial power shape	A
Axial power shape	Uniform
Number of MV spacers	5
Number of NMV spacers	2
Number of simple spacers	6
MV spacer location (mm)	610, 1219, 1829, 2438, 3048
NMV spacer location (mm)	0, 3658
Simple spacer location (mm)	305, 914, 1524, 2134, 2743, 3353

○ : Heated rod ◍ : Thimble rod
MV: Mixing vane, NMV: No mixing vane
Spacer location is distance from bottom of heated length to spacer bottom face.

Table 3.2-3 Geometry and power shape of test assembly A1, A2 and A3

Item	Data		
Assembly	 **A1**	 **A2**	 **A3**
Rods array	5×5	5×5	6×6
Number of heated rods	25	25	36
Number of thimble rods	0	0	0
Heated rod outer diameter (mm)	9.50	9.50	9.50
Thimble rod outer diameter (mm)	-	-	-
Heated rods pitch (mm)	12.60	12.60	12.60
Axial heated length (mm)	3658	3658	3658
Flow channel inner width (mm)	64.9	64.9	77.5
Radial power shape	C	A	D
Axial power shape	Uniform	Uniform	Uniform
Number of MV spacers	7	7	7

Number of NMV spacer	2	2	2
Number of simple spacers	8	8	8
MV spacer location (mm)	457, 914, 1372, 1829, 2286, 2743, 3200		
NMV spacer location (mm)	0, 3658		
Simple spacer location (mm)	229, 686, 1143, 1600, 2057, 2515, 2972, 3429		

Table 3.2-4 Geometry and power shape of test assembly A4 and A8

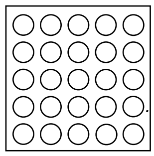
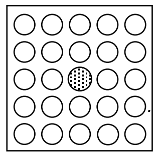
Item	Data	
Assembly	 **A4, A11**	 **A8, A12**
Rods array	5×5	5×5
Number of heated rods	25	24
Number of thimble rods	0	1
Heated rod outer diameter (mm)	9.50	9.50
Thimble rod outer diameter (mm)	-	12.24
Heated rods pitch (mm)	12.60	12.60
Axial heated length (mm)	3658	3658
Flow channel inner width (mm)	64.9	64.9
Radial power shape	A	B
Axial power shape	Cosine	Cosine
Number of MV spacers	7	7
Number of NMV spacer	2	2
Number of simple spacers	8	8
MV spacer location (mm)	471, 925, 1378, 1832, 2285, 2739, 3247	
NMV spacer location (mm)	2.5, 3755	
Simple spacer location (mm)	237, 698, 1151, 1605, 2059, 2512, 2993, 3501	

Table 2.2-4 Tolerance of manufacturing for test assembly

Item	Tolerance
Rod bundle	
Heater rod diameter	0.02 mm
Heater rod displacement	0.45 mm
Flow channel inner width	0.05 mm
Flow channel displacement	0.20 mm
Power distribution	3 %

Table 3.2-6 Radial power distribution

Pattern A

0.85	0.85	0.85	0.85	0.85
0.85	1	1	1	0.85
0.85	1	1	1	0.85
0.85	1	1	1	0.85
0.85	0.85	0.85	0.85	0.85

Pattern B

0.85	0.85	0.85	0.85	0.85
0.85	1	1	1	0.85
0.85	1	0	1	0.85
0.85	1	1	1	0.85
0.85	0.85	0.85	0.85	0.85

Pattern C

1	1	0.25	0.25	0.25
1	1	1	0.25	0.25
1	1	0.25	0.25	0.25
1	1	1	0.25	0.25
1	1	0.25	0.25	0.25

Pattern D

0.85	0.85	0.85	0.85	0.85	0.85
0.85	1	1	1	1	0.85
0.85	1	1	1	1	0.85
0.85	1	1	1	1	0.85
0.85	1	1	1	1	0.85
0.85	0.85	0.85	0.85	0.85	0.85

Table 3.2-7 Axial power shape

Node	Relative power
	Cosine
(Bottom)	
1	0.42
2	0.47
3	0.56
4	0.67
5	0.80
6	0.94
7	1.08
8	1.22
9	1.34
10	1.44
11	1.51
12	1.55
13	1.55
14	1.51
15	1.44
16	1.34
17	1.22
18	1.08
19	0.94
20	0.80
21	0.67
22	0.56
23	0.47
24	0.42
(Top)	

(2) Heater rod structure

Heater rod structure is specified in Table 3.2-8.

Table 3.2-8 Heater rod structure for bundle test assembly

Item		Data
Heater	Outer diameter (mm)	9.5
	Thickness (mm)	0.65
	Material	Inconel 600
	Heating method	Direct heating
Insulator	Outer diameter (mm)	8.2
	Inner diameter (mm)	5.8
	Material	Alumina

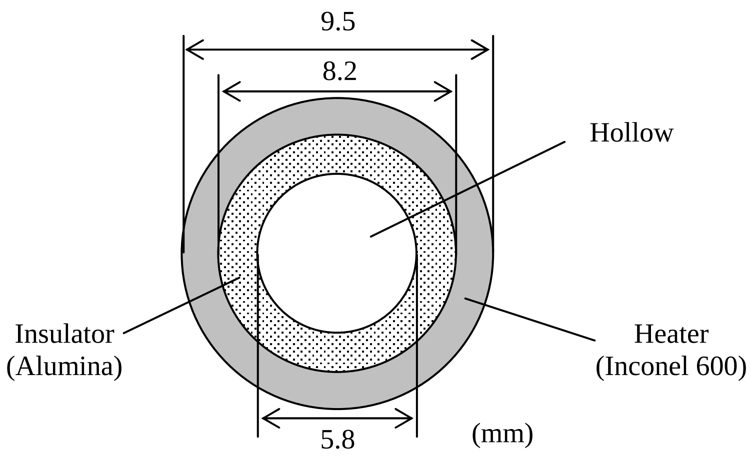


Figure 3.2-1 Cross-sectional view of heater rod for bundle test assembly

(3) Spacer data

Table 3.2-9 Spacer geometry data

Item		Data
Spacer width (mm)		64.5
Spacer heights (mm)	Simple spacer	12.8
	MV and NMV spacer	43.6

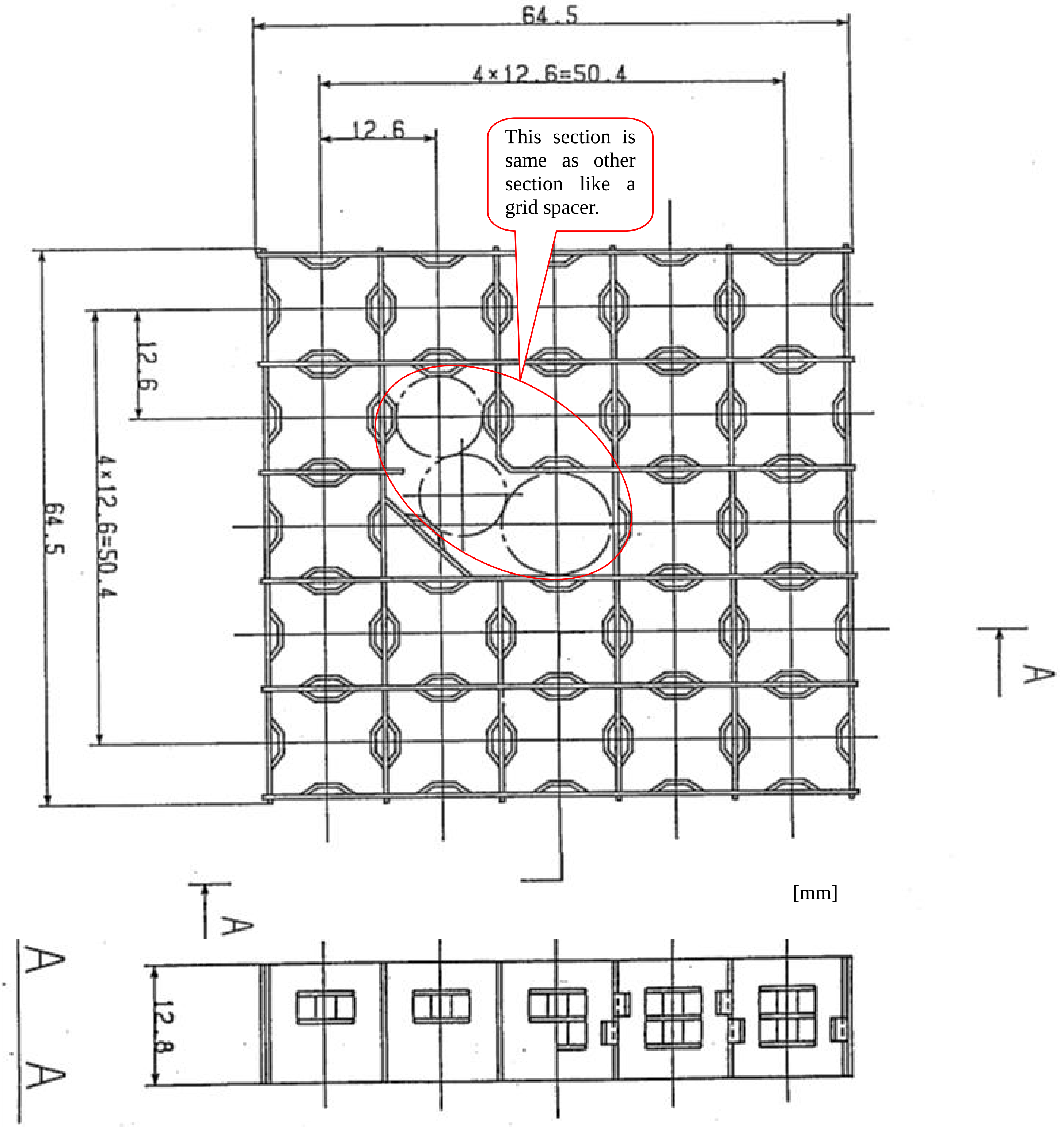


Figure 3.2-2 Schematic of simple spacer

(3) Location of thermocouples

The rod surface temperature was measured at several locations as depicted in Figure 3.2-5.

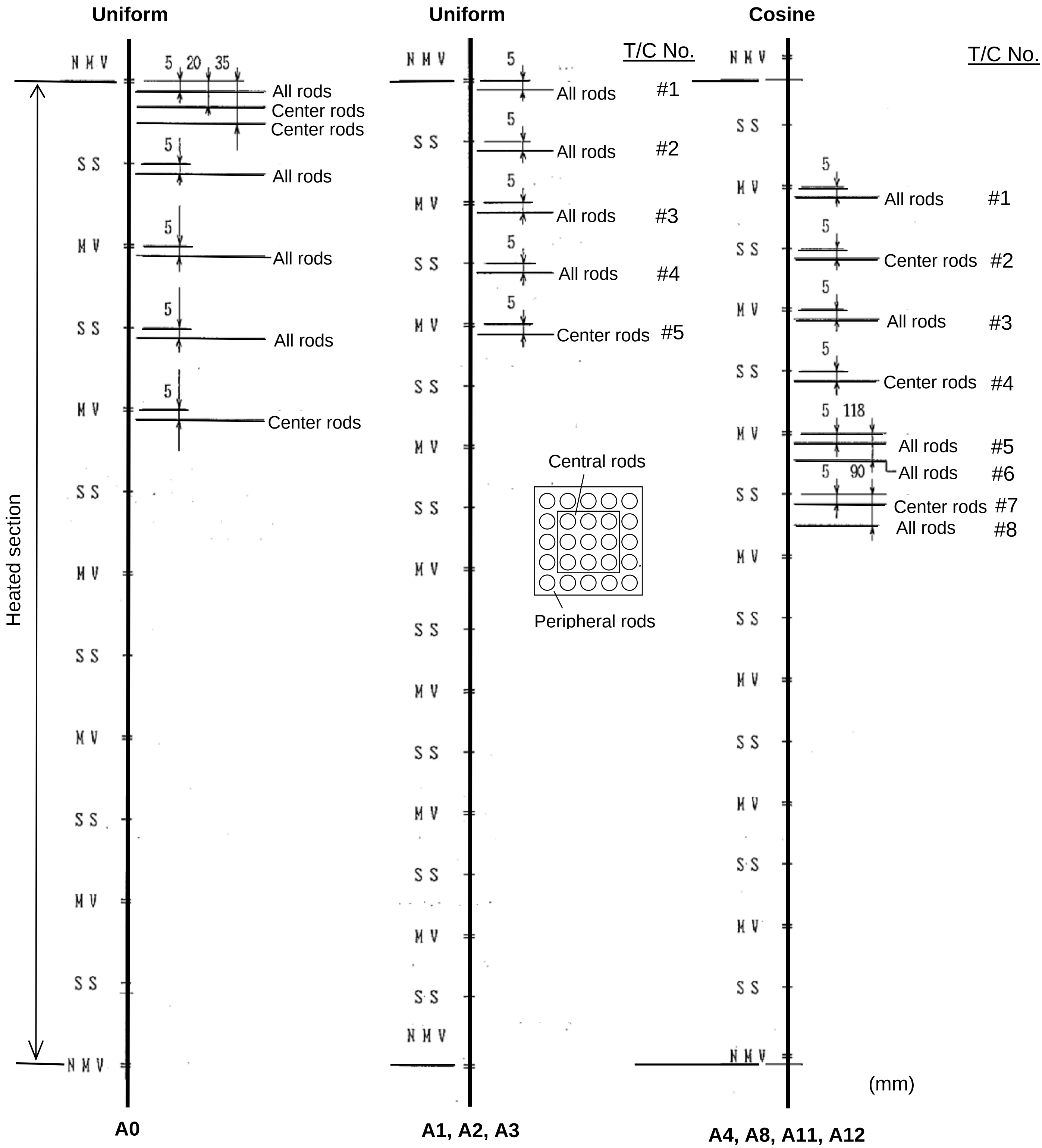


Figure 3.2-5 Location of thermocouples

(4) Fluid temperature measurements

Fluid temperature was measured in each subchannel at 457 mm upper from the top end of the heated section in test assembly A1 as depicted in Figure 3.2-6.

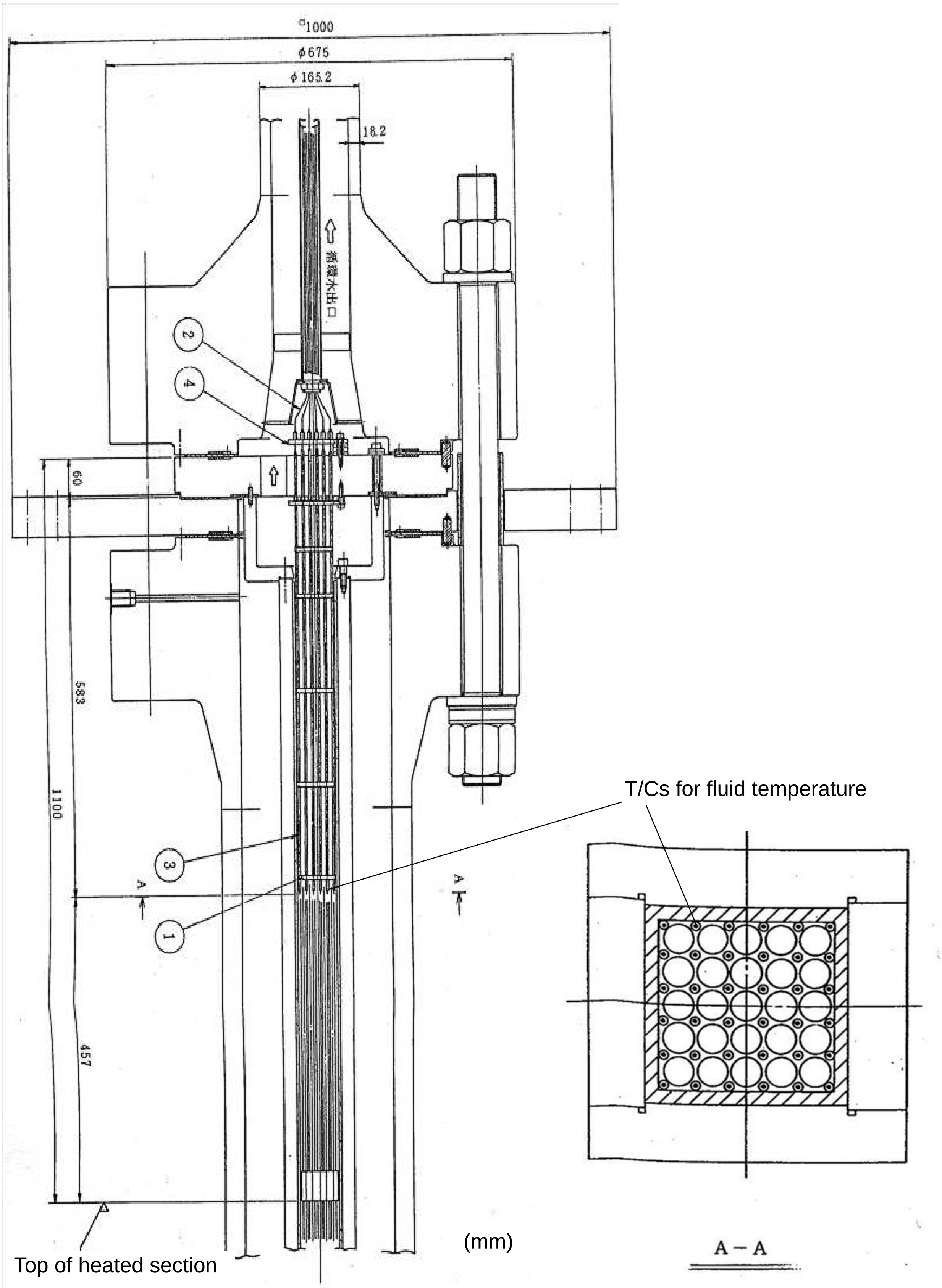


Figure 3.2-6 Fluid temperature measurements in test assembly A1

4. Benchmark Database

Much data has been accumulated by NUPEC PWR subchannel and bundle test series. NUPEC PSBT benchmark database is summarized in Table 4-1. NUPEC PSBT data would be opened for the benchmark participants with a propriety control agreement (Appendix A). Samples of PSBT data is shown in Appendix B.

Table 4-1 NUPEC PSBT benchmark database

Items of data
Void fraction measurements - Steady-state void fraction in subchannel - Steady-state void fraction in rod bundle - Transient void fraction in rod bundle
DNB measurements - Steady-state DNB data in rod bundle - Steady-state fluid temperature distribution in rod bundle - Transient DNB data in rod bundle

4.1 Void fraction measurements

(1) Definition of test series

Table 4.1-1 summarizes test series for void fraction measurement.
Void fraction measurement was conducted with DNB condition in some test case of test series 5, 6 and 7.
A part of test case of test series 5 was duplicated in test series 8.

Table 4.1-1 Test series for void fraction measurement

Test series	Test section	Assembly	Test mode		Void measurement	
			Steady-state	Transient	CT	Chordal
1	Subchannel	S1	Y		Y	Y
2		S2	Y		Y	Y
3		S3	Y		Y	Y
4		S4	Y		Y	Y
5	5×5 Rod bundle	B5	Y			Y
5T				Y		Y
6		B6	Y			Y
6T				Y		Y
7		B7	Y			Y
7T				Y		Y
8		B5	Y			Y

Table 4.1-2 Reference rated operating condition of PWR

Test section	Pressure (kg/cm^2a)	Mass flux ($10^6 kg/m^2h$)	Heat flux ($10^6 kcal/m^2h$)	Power (MW)	Inlet temperature (Celsius)
Subchannel	158	12	0.5	N/A	290
5×5 Rod bundle	158	12	0.5	2.7	290

(2) Test matrix

Table 4.1-3 Test matrix for steady-state void measurement test series 1

Pressure (kg/cm^2a)	Mass flux ($10^6 kg/m^2h$)	Power (kW)	Heat flux ($10^6 kcal/m^2h$)	Inlet temperature (Celsius)
169	15	60	0.96	335
	11	80	1.28	315
	11	50	0.80	$\underline{330}$ $\underline{335}$ $\underline{340}$
	5	40	0.64	335
150	15	100	1.60	305
	15	80	1.28	$\underline{310}$ $\underline{320}$
	11	90	1.44	$\underline{295}$ $\underline{300}$
	11	70	1.12	$\underline{300}$ 310 $\underline{320}$
	11	60	0.96	$\underline{300}$ 305 $\underline{310}$ $\underline{315}$
				$\underline{320}$ $\underline{325}$ $\underline{330}$
	8	50	0.80	315
	5	80	1.28	250
	5	60	0.96	$\underline{270}$ $\underline{285}$ $\underline{300}$
	5	20	0.32	320
125	15	70	1.12	315
	11	90	1.44	280
	11	60	0.96	$\underline{295}$ $\underline{310}$ $\underline{320}$
	5	40	0.64	295
100	11	100	1.60	260
	11	70	1.12	$\underline{275}$ $\underline{305}$
	8	60	0.96	300
	5	80	1.28	$\underline{215}$ $\underline{250}$
	5	60	0.96	$\underline{210}$ 220 $\underline{230}$ $\underline{240}$
				$\underline{255}$ $\underline{270}$ $\underline{290}$
	2	20	0.32	$\underline{255}$ $\underline{285}$
75	11	80	1.28	260
	5	90	1.44	165
	5	50	0.80	$\underline{220}$ $\underline{245}$ $\underline{265}$
	2	20	0.32	245
50	8	70	1.12	215
	5	80	1.28	$\underline{165}$ $\underline{200}$
	5	50	0.80	$\underline{190}$ $\underline{205}$ $\underline{240}$
	2	20	0.32	$\underline{205}$ $\underline{240}$

$\underline{\quad\quad}$: CT measurement case

Table 4.1-4 Test matrix for steady-state void measurement test series 2

Pressure (kg/cm^2a)	Mass flux ($10^6 kg/m^2h$)	Power (kW)	Heat flux ($10^6 kcal/m^2h$)	Inlet temperature (Celsius)
169	15	45	0.96	340
	11	52.5	1.12	330
	11	37.5	0.80	<u>335</u> <u>340</u> <u>345</u>
	5	30	0.64	345
150	15	67.5	1.44	315
	15	60	1.28	<u>320</u> <u>325</u>
	11	60	1.28	<u>310</u> <u>315</u>
	11	52.5	1.12	<u>310</u> 320 <u>330</u>
	11	45	0.96	<u>305</u> 310 <u>315</u> <u>320</u>
				<u>325</u> <u>330</u> <u>335</u>
	8	37.5	0.80	325
	5	60	1.28	280
	5	45	0.96	<u>290</u> <u>305</u> <u>320</u>
	5	15	0.32	325
125	15	52.5	1.12	315
	11	67.5	1.44	295
	11	45	0.96	<u>305</u> <u>310</u> <u>320</u>
	5	30	0.64	300
100	11	75	1.60	270
	11	52.5	1.12	<u>280</u> <u>300</u>
	8	45	0.96	300
	5	60	1.28	<u>245</u> <u>280</u>
	5	45	0.96	<u>235</u> 245 <u>255</u> <u>265</u>
				<u>275</u> <u>285</u> <u>300</u>
	2	15	0.32	<u>275</u> <u>295</u>
75	11	60	1.28	275
	5	67.5	1.44	200
	5	37.5	0.80	<u>240</u> <u>265</u> <u>280</u>
	2	15	0.32	265
50	8	52.5	1.12	230
	5	60	1.28	<u>195</u> <u>230</u>
	5	37.5	0.80	<u>210</u> <u>225</u> <u>255</u>
	2	15	0.32	<u>225</u> <u>255</u>

___: CT measurement case

Table 4.1-5 Test matrix for steady-state void measurement test series 3

Pressure (kg/cm^2a)	Mass flux ($10^6 kg/m^2h$)	Power (kW)	Heat flux ($10^6 kcal/m^2h$)	Inlet temperature (Celsius)
169	15	30	0.96	340
	11	35	1.12	325
	11	25	0.80	335 340 345
	5	20	0.64	345
150	15	45	1.44	315
	15	40	1.28	315 325
	11	40	1.28	310 315
	11	35	1.12	310 320 325
	11	30	0.96	305 310 315 320
				325 330 335
	8	25	0.80	320
	5	40	1.28	275
	5	30	0.96	285 300 315
	5	10	0.32	325
125	15	35	1.12	320
	11	45	1.44	285
	11	30	0.96	300 310 320
	5	20	0.64	300
100	11	50	1.60	275
	11	35	1.12	285 295
	8	30	0.96	300
	5	40	1.28	240 275
	5	30	0.96	230 240 250 260
				270 285 300
	2	10	0.32	270 300
75	11	40	1.28	270
	5	45	1.44	195
	5	25	0.80	235 260 280
	2	10	0.32	260
50	8	35	1.12	230
	5	40	1.28	190 225
	5	25	0.80	205 220 255
	2	10	0.32	220 255

___: CT measurement case

Table 4.1-6 Test matrix for steady-state void measurement test series 4

Pressure (kg/cm^2a)	Mass flux ($10^6kg/m^2h$)	Power (kW)	Heat flux ($10^6kcal/m^2h$)	Inlet temperature (Celsius)
169	15	15	0.96	340
	11	17.5	1.12	330
	11	12.5	0.80	335 340 345
	5	10	0.64	345
150	15	22.5	1.44	320
	15	20	1.28	320 330
	11	20	1.28	315 320
	11	17.5	1.12	315 320 330
	11	15	0.96	311 315 319 323 327 331 335
	8	12.5	0.80	325
	5	20	1.28	290
	5	15	0.96	295 310 325
	5	10	0.64	325
125	15	17.5	1.12	320
	11	22.5	1.44	300
	11	15	0.96	305 310 320
	5	10	0.64	310
100	11	25	1.60	280
	11	17.5	1.12	290 300
	8	15	0.96	295
	5	20	1.28	255 285
	5	15	0.96	240 250 260 270 280 290 300
	2	5	0.32	280 300
75	11	20	1.28	275
	5	22.5	1.44	215
	5	12.5	0.80	245 265 280
	2	5	0.32	270
50	8	17.5	1.12	235
	5	20	1.28	205 240
	5	12.5	0.80	215 230 255
	2	5	0.32	230 255

___: CT measurement case

Table 4.1-7 Test matrix for steady-state void measurement test series 5

Pressure (kg/cm^2a)	Mass flux ($10^6 kg/m^2h$)	Power (MW)	Heat flux ($10^6 kcal/m^2h$)	Inlet temperature (Celsius)	
169	15	3.0	1.05	315	320
	11	3.0	1.05	290	295
	11	2.5	0.87	310	315
	8	2.5	0.87	285	290
	8	2.0	0.70	310	315
	5	1.5	0.52	300	310
150	15	3.5	1.22	290	295
	15	2.5	0.87	315	320
	11	3.0	1.05	285	**290**
	11	2.5	0.87	300	310
	11	2.0	0.70	315	320
	8	2.5	0.87	275	**285**
	8	2.0	0.70	300	310
	8	1.5	0.52	315	320
	5	2.0	0.70	255	260
	5	1.0	0.35	310	320
125	15	3.5	1.22	275	280
	11	3.0	1.05	270	275
	8	3.0	1.05	240	245
	8	2.5	0.87	265	275
	5	2.0	0.70	245	255
100	11	3.5	1.22	240	245
	8	3.0	1.05	235	240
	5	2.5	0.87	195	**200**
	5	2.0	0.70	220	230
	2	1.0	0.35	200	210
75	11	4.0	1.39	205	210
	8	3.5	1.22	190	200
	8	3.0	1.05	200	215
	5	2.5	0.87	170	180
	5	2.0	0.70	200	210
	2	1.0	0.35	180	190
50	8	3.5	1.22	160	170
	8	3.0	1.05	170	180
	5	2.5	0.87	145	155
	5	2.0	0.70	165	175
	2	1.0	0.35	145	155

Bold: with DNB case

Table 4.1-8 Test matrix for steady-state void measurement test series 6

Pressure (kg/cm^2a)	Mass flux ($10^6 kg/m^2h$)	Power (MW)	Heat flux ($10^6 kcal/m^2h$)	Inlet temperature (Celsius)
169	15	3.5	1.22	305 310
	11	3.5	1.22	280 285
	11	3.0	1.05	300 305
	8	3.0	1.05	265 270
	8	2.5	0.87	290 295
	5	2.0	0.70	270 275
150	15	4.0	1.39	280 285
	15	3.0	1.05	305 315
	11	3.5	1.22	270 **275**
	11	3.0	1.05	285 295
	11	2.5	0.87	305 315
	8	3.0	1.05	255 **265**
	8	2.5	0.87	280 290
	8	2.0	0.70	305 315
	5	2.5	0.87	220 230
	5	1.0	0.35	310 320
125	15	3.5	1.22	275 285
	11	3.0	1.05	270 280
	8	3.0	1.05	240 250
	8	2.5	0.87	265 280
	5	2.0	0.70	240 260
100	11	3.5	1.22	225 240
	8	3.0	1.05	220 240
	5	2.5	0.87	180 **200**
	5	2.0	0.70	210 230
	2	1.0	0.35	190 210
75	11	4.0	1.39	195 210
	8	3.5	1.22	170 190
	8	3.0	1.05	185 200
	5	2.5	0.87	150 170
	5	2.0	0.70	180 200
	2	1.0	0.35	160 175
50	8	3.5	1.22	150 165
	8	3.0	1.05	160 175
	5	2.5	0.87	140 155
	5	2.0	0.70	160 175
	2	1.0	0.35	140 155

Bold: with DNB case

Table 4.1-9 Test matrix for steady-state void measurement test series 7

Pressure (kg/cm^2a)	Mass flux ($10^6kg/m^2h$)	Power (MW)	Heat flux ($10^6kcal/m^2h$)	Inlet temperature (Celsius)	
169	15	3.5	1.28	305	310
	11	3.5	1.28	280	285
	11	3.0	1.09	300	305
	8	3.0	1.09	265	270
	8	2.5	0.91	290	295
	5	2.0	0.73	270	275
150	15	4.0	1.46	280	285
	15	3.0	1.09	305	315
	11	3.5	1.28	270	**275**
	11	3.0	1.09	285	295
	11	2.5	0.91	305	315
	8	3.0	1.09	255	**265**
	8	2.5	0.91	280	290
	8	2.0	0.73	305	315
	5	2.5	0.91	220	230
	5	1.0	0.36	310	320
125	15	3.5	1.28	275	285
	11	3.0	1.09	270	280
	8	3.0	1.09	240	250
	8	2.5	0.91	265	280
	5	2.0	0.73	240	260
100	11	3.5	1.28	225	240
	8	3.0	1.09	220	240
	5	2.5	0.91	180	**200**
	5	2.0	0.73	210	230
	2	1.0	0.36	190	210
75	11	4.0	1.46	195	210
	8	3.5	1.28	170	190
	8	3.0	1.09	185	200
	5	2.5	0.91	150	170
	5	2.0	0.73	180	200
	2	1.0	0.36	160	175
50	8	3.5	1.28	150	165
	8	3.0	1.09	160	175
	5	2.5	0.91	140	155
	5	2.0	0.73	160	175
	2	1.0	0.36	140	155

Bold: with DNB case

Table 4.1-10 Test matrix for transient void measurement test series 5T, 6T and 7T

Test series	Assembly	Initial Conditions				Transients
		Pressure (kg/cm^2a)	Mass flux ($10^6 kg/m^2h$)	Power (MW)	Inlet temperature (Celsius)	
5T	B1	158	12	2.25	300	Power increase Flow reduction Depressurization Temperature increase
6T	B2	158	12	2.6	290	Power increase Flow reduction Depressurization Temperature increase
7T	B3	158	12	2.5	290	Power increase Flow reduction Depressurization Temperature increase

4.2 DNB Measurements

(1) Definition of test series

Table 4.2-1 summarizes test series for DNB measurements.
Fluid temperature was measured in each subchannel at 457 mm upper from the top end of the heated section in test assembly A1 without DNB condition in test series 1.

Table 4.2-1 Test series for DNB measurement

Test series	Test section	Assembly	Test mode		Measurement	
			Steady-state	Transient	DNB	Fluid temperature
0	5×5	A0	Y		Y	
1		A1	Y			Y
2		A2	Y		Y	
3	6×6	A3	Y		Y	
4	5×5	A4	Y		Y	
8		A8	Y		Y	
11T		A11		Y	Y	
12T		A12		Y	Y	
13		A4	Y		Y	

Table 4.2-2 Reference rated operating condition of PWR

Test section	Pressure (kg/cm^2a)	Mass flux ($10^6 kg/m^2h$)	Heat flux ($10^6 kcal/m^2h$)	Power (MW)	Inlet temperature (Celsius)
5×5 Rod bundle	158	12	0.5	2.7	290
6×6 Rod bundle	158	12	0.5	3.9	290

(2) Test matrix

Table 4.2-3 Test matrix for steady-state DNB test series 0

Inlet subcooling (kcal/kg)

Pressure (kg/cm^2a)	**Mass flux ($\times 10^6 kg/m^2h$)**							
	0.5	1.2	2	5	8	11	14	17
100	-	-	-	85 155	85 155	30 120 190	85 155	85 140
125	-	-	-	85 155	85 155	30 120 190	85 155	85 140
150	-	-	-	30 120 155 190	30 120 155 190	30 85 120 190	30 120 155 190	30 120 150
169	-	-	-	85 155	85 155	30 120 190	85 155	85 140

Referenced rated condition: Pressure=158, Mass flux =12.1, Inlet subcooling=82.6

Table 4.2-4 Test matrix for steady-state DNB test series 1

Outlet quality (%)

Pressure (kg/cm^2a)	**Enthalpy increase (kcal/kg)**	**Mass flux ($\times 10^6 kg/m^2h$)**					
		0.5	1.2	2	5	11	17
50	70	-30	-30	-30	-30	-30	-30
100	70	-30	-30	-30	-30	-5 -10 -30 -50	-30
150	50				-10 -30 -50 -80	-10 -30 -50 -80	
	70	-30	-30	-5 -10 -30 -50 -80	-5 -10 -30 -50 -80	-5 -10 -30 -50 -80	-30
	90			-10 -30 -50 -80	-10 -30 -50 -80	-10 -30 -50 -80	
169	70	-30	-30	-30	-30	-30	-30

Table 4.2-5 Test matrix for steady-state DNB test series 2

Inlet subcooling (kcal/kg)

Pressure (kg/cm^2a)	**Mass flux ($\times 10^6 kg/m^2h$)**							
	0.5	1.2	2	5	8	11	14	17
50	85 155	85 155	85 155	85 155	85 155	30 85 155	85 120	85
75	-	-	-	85 155		30 85 155		85
100	85 155	85 155	85 155	85 155	85 155	30 120 155 190	85 155	85
125	-	-	-	85 155	-	30 120 190	-	85 120
150	85 155	85 155	85 155	30 120 190	30 120 190	30 85 120 155 190	30 120 155	30 85 120
169	120	-	-	120	-	30 85 120 155 190	-	120

Referenced rated condition: Pressure=158, Mass flux =12.1, Inlet subcooling=82.6

Table 4.2-6 Test matrix for steady-state DNB test series 3

Inlet subcooling (kcal/kg)

Pressure (kg/cm^2a)	**Mass flux ($\times 10^6 kg/m^2h$)**							
	0.5	1.2	2	5	8	11	14	17
100	-	-	-	85 155	85 155	30 120 190	85 120	85
125	-	-	-	85 155	85 155	30 120 190	85 120	85 120
150	-	-	-	30 120 190	30 120 190	30 85 120 155 190	30 120 155	30 85 120
169	-	-	-	85 120 155	85 155	30 120 155	85 155	85 120

Referenced rated condition: Pressure=158, Mass flux =12.1, Inlet subcooling=82.6

Table 4.2-7 Test matrix for steady-state DNB test series 4

Inlet subcooling (kcal/kg)

Pressure (kg/cm^2a)	**Mass flux ($\times 10^6 kg/m^2h$)**							
	0.5	1.2	2	5	8	11	14	17
50	-	-	85 120	60 85	-	-	-	-
75	-	-	85 120	60 85 120	60	30 60 85	-	-
100	-	-	85 120	60 85 120	85	30 60 85 120	30 60 85	60
125	-	-	85 120	60 85 120 155	-	30 60 85 120	60 85	85
150	-	-	85 120	30 60 85 120 155	30 85 155	30 60 85 120	30 60 85 120	30 85
169	-	-	-	85 120 155 190	-	30 60 85 120 155	-	30 60 85

Referenced rated condition: Pressure=158, Mass flux =12.1, Inlet subcooling=82.6

Table 4.2-8 Test matrix for steady-state DNB test series 8

Inlet subcooling (kcal/kg)

Pressure (kg/cm^2a)	**Mass flux ($\times10^6 kg/m^2h$)**							
	0.5	1.2	2	5	8	11	14	17
50	-	-	85 120	60 85 120	-	30 60 85	-	-
75	-	-	85 120	60 85 120	60	30 60 85	-	-
100	-	-	85 120	60 85 120	30 60 85	30 60 85 120	30 60 85	60
125	-	-	85 120	60 85 120 155	30 60 85	30 60 85 120	60 85	85
150	-	-	85 120	30 60 85 120 155	30 60 85 120 155	30 60 85 120	30 60 85 120	30 85
169	-	-	-	85 120 155 190	85 120	30 60 85 120 155	-	30 60 85

Referenced rated condition: Pressure=158, Mass flux =12.1, Inlet subcooling=82.6

Table 4.2-9 Test matrix for steady-state DNB test series 13

Inlet subcooling (kcal/kg)

Pressure (kg/cm^2a)	Mass flux ($\times 10^6 kg/m^2h$)							
	5	6.5	8	9.5	11	12.5	14	17
125	-	-	85	-	85	-	-	-
137.5	-	-	85	-	85	-	-	-
159.5	-	-	30 60 72.5 85 102.5 120	30 85	30 60 72.5 85 102.5 120	30 85	30 85	-
150	-	-	85	-	85	-	-	-
169	-	-	85	-	85	-	-	-

Referenced rated condition: Pressure=158, Mass flux =12.1, Inlet subcooling=82.6

Table 4.2-10 Test matrix for transient DNB test series 11T and 12T

No.	Assembly	Initial Conditions				Transients
		Power (MW)	Mass flux ($\times 10^6 kg/m^2h$)	Pressure (kg/cm^2a)	Inlet temperature (Celsius)	
11T	A4	2.7	12	158	290	Power increase Flow reduction Depressurization Temperature increase
12T	A8	2.7	12	158	290	Power increase Flow reduction Depressurization Temperature increase

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[2] Hori, K. et al., In Bundle Void Fraction Measurement of PWR Fuel Assembly, *ICONE-2*, Vol.1, pp.69-76, San Francisco, California, 21-24 March 1993.

[3] Hori, K. et al., Void Fraction in A Single Channel Simulating One Subchannel of A PWR Fuel Assembly, *Two-Phase Flow Modeling and Experimentation*, pp.1013-1027, 1995.

[4] Akiyama, M. et al., Reliability Proving Test on Maximum Thermal Loading of PWR Fuel Assembly, *Journal of Nuclear Science and Technology*, Vol. 36, No. 1, p. 47, 1994 (in Japanese).

[5] Pamphlet of Takasago Engineering Laboratory, Nuclear Power Engineering Center (NUPEC).

APPENDIX A

Conditions for Release, Rules and Restrictions Applying to the NUPEC PWR Subchannel and Bundle Test Experimental Data and Information

The data and other experimental information are released under the following conditions:

- The data and information remain the property of the Japan Nuclear Safety Organization (JNES).
- The data and information is exclusively released to participants in the OECD/NRC PSBT Benchmark Study
- Recipients of the data will use it exclusively for model and code improvements in the two-phase flow area and agree to provide in return details of the methods and models used in the interpretation / simulation of the experiments, including results of sensitivity analysis
- Recipients agree not to make copies of the information and not to further distribute or sell it to third parties
- Recipients agree to provide feedback on deficiencies or errors they may find in the data.
- Recipients agree to inform the benchmark organizers before publishing any study results for which the PSBT benchmark data have been used.

DISCLAIMER
Neither the Organization for Economic Co-operation and Development nor JNES, nor any person acting on behalf of any of these organisations assume any liability with respect to the use of, or for damage resulting from the use of the data and information released.

I agree to these conditions:

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Signature:

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Computer Program Service
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Fax: +33 1 4524 1110

APPENDIX B

Sample of NUPEC PWR Subchannel and Bundle Test Benchmark Data

Table B-1 List of PSBT benchmark database

Items of data	Test series	Vol.II Part
Void fraction measurements data		
- Steady-state void fraction in subchannel by CT measurement	1, 2, 3, 4	1
- Steady-state void distribution image in subchannel by CT measurement	1, 2	2
- Steady-state void fraction in rod bundle by chordal measurement	5, 6, 7, 8	1
- Steady-state void distribution image in rod bundle by chordal measurement	5, 6, 7, 8	3
- Transient void fraction in rod bundle by chordal measurement	5T, 6T, 7T	1
DNB measurements data		
- Steady-state DNB data in rod bundle	0, 2, 3, 4, 8, 13	4
- Steady-state DNB detected location in rod bundle	4, 8, 13	4
- Steady-state fluid temperature distribution in rod bundle	1	4
- Transient DNB data in rod bundle	11T, 12T	4

B.1 Void Fraction Measurements

Table B.1-1 Sample of steady-state void fraction in subchannel by CT measurement in test series 1

Run No	Pressure (ata)	Mass flux (x10^{6}kg/m^{2}h)	Power (kw)	Inlet temperature (℃)	Subchannel averaged Fluid density (Kg/m^3)	Subchannel averaged Void fraction
1.1221	169.1	10.95	49.9	329.7	554.	0.087
1.1222	169.1	10.98	50.0	334.7	517.	0.142
1.1223	169.1	11.00	49.9	339.7	434.	0.332
1.4121	100.1	10.97	69.9	274.1	635.	0.097
1.4122	99.8	10.90	69.8	304.5	286.	0.636
1.4311	100.4	5.01	79.9	214.2	563.	0.215
1.4312	100.2	5.03	79.8	248.9	331.	0.566
1.4321	100.5	5.01	59.9	209.3	718.	0.045
1.4323	100.5	5.03	59.9	229.2	684.	0.047
1.4324	100.1	5.02	60.1	238.9	600.	0.157

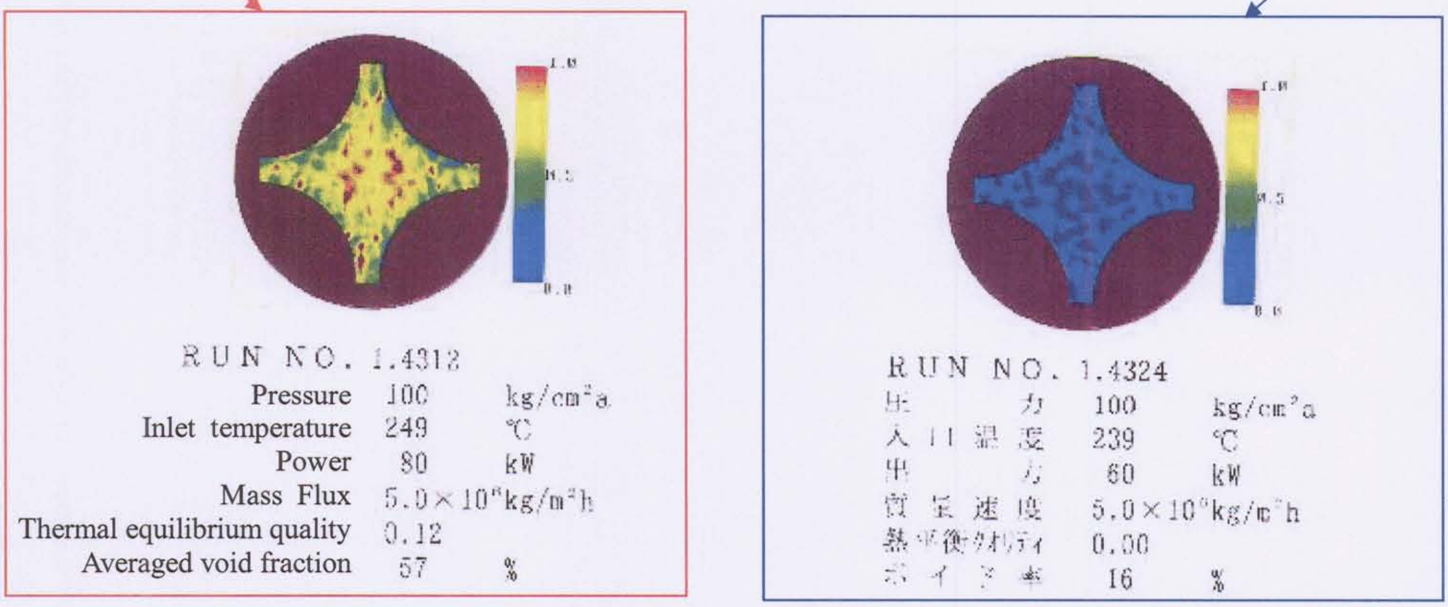
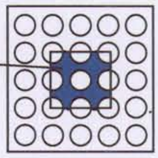


Figure B.1-1 Sample of void distribution image in subchannel by CT measurement in test series 1

Table B.1-2 Sample of steady-state void fraction in rod bundle by chordal measurement in test series 5

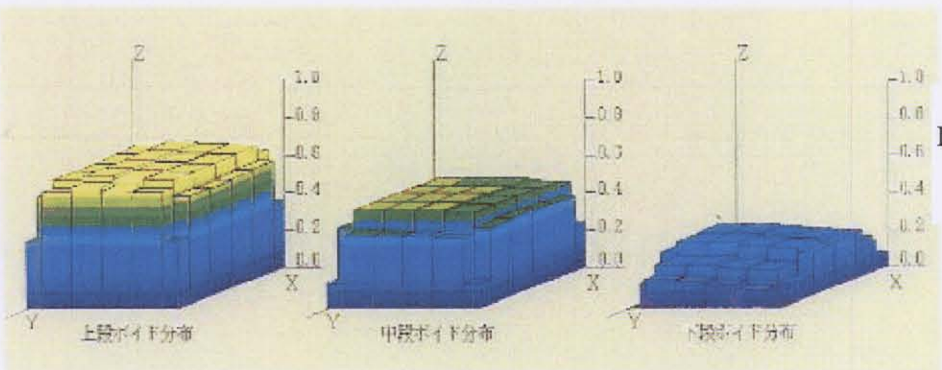
Run No	Pressure (ata)	Mass flux ($x10^6 kg/m^2h$)	Power (kw)	Inlet temperature (℃)	Central regional 4 subchannel averaged void fraction		
					Lower	Middle	Upper
5.1121	167.39	14.96	2990.	316.9	0.0000	0.0000	0.1791
5.1122	167.25	15.02	2995.	322.0	0.0000	0.0907	0.2996
5.1221	168.29	11.00	3000.	292.3	0.0000	0.0000	0.0000
5.1222	168.27	10.98	2998.	297.3	0.0000	0.0000	0.0769
5.1231	168.29	10.97	2488.	311.8	0.0000	0.0000	0.1951
5.1232	168.29	10.98	2486.	316.6	0.0000	0.0794	0.3056
5.1331	168.80	7.95	2491.	287.2	0.0000	0.0000	0.1153
5.1332	168.71	7.93	2491.	292.2	0.0000	0.0000	0.2519
5.1341	168.76	7.90	2005.	312.2	0.0000	0.0981	0.3439
5.1342	168.83	7.91	2005.	317.2	0.0700	0.2304	0.4147
5.1451	169.19	4.99	1515.	302.3	0.0000	0.0779	0.3393
5.1452	169.19	4.98	1516.	312.5	0.1233	0.2844	0.4709
5.5432	75.18	5.05	2518.	184.2	0.1110	0.4116	0.6641
5.5441	75.13	5.01	2020.	203.6	0.0452	0.3300	0.5737
5.5442	75.27	5.02	2022.	213.6	0.2283	0.4497	0.6652
5.5551	75.41	2.00	1028.	183.0	0.0540	0.3473	0.5991
5.5552	75.40	2.00	1026.	193.2	0.2226	0.4631	0.6684

Central regional 4 subchannel



Central regional 4 subchannel averaged void fraction

Upper 67 % Middle 46 % Lower 22 %



RUN NO. 5.5552
Pressure 75 kg/cm^2a
Inlet temperature 193 ℃
Power 1026 kW
Mass flux $2.0\times10^6 kg/m^2h$
TE quality U 0.15
M 0.07
L 0.00

Figure B.1-2 Sample of void distribution image in rod bundle by chordal measurement in test series 5

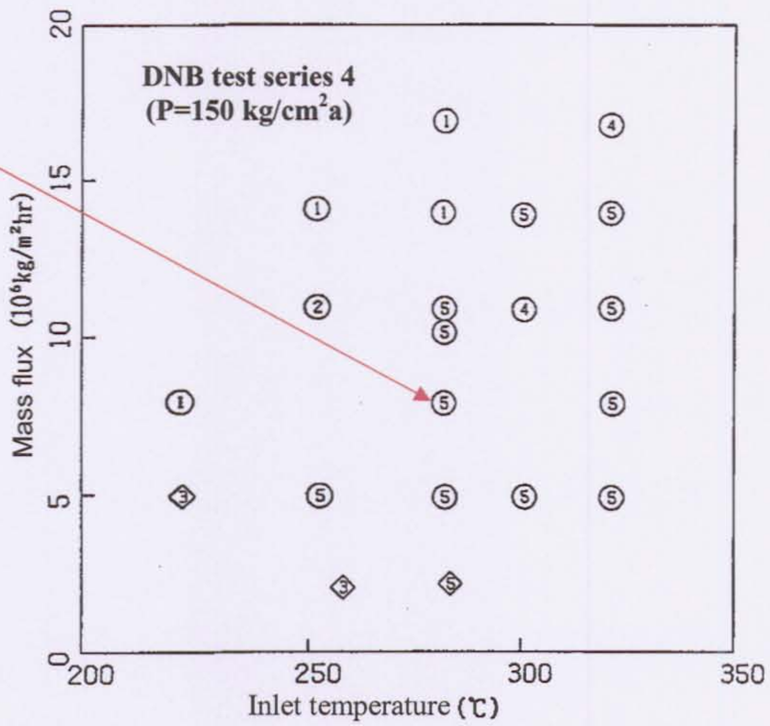
Table B.1-3 Sample of transient void fraction in rod bundle by chordal measurement in test series 5T

Time sec	Pressure kg/cm²a	Mass flux x10⁶kg/m²h	Power KW	Inlet temperature ℃	Central regional 4 subchannel averaged void fraction Lower	Middle	Upper
30.0	154.2	11.95	2282.	300.4	0.0000	0.0000	0.0440
30.2	154.4	11.95	2261.	300.4	0.0124	0.0000	0.0465
30.4	154.3	11.94	2254.	300.4	0.0000	0.0000	0.0379
30.6	153.9	11.96	2256.	300.4	0.0000	0.0000	0.0225
30.8	154.1	11.96	2261.	300.4	0.0000	0.0000	0.0052
31.0	154.0	11.95	2182.	300.5	0.0000	0.0000	0.0053
31.2	154.1	11.95	2194.	300.4	0.0000	0.0041	0.0256
31.4	154.2	12.01	2228.	300.6	0.0109	0.0000	0.0357
31.6	154.3	12.00	2236.	300.5	0.0205	0.0000	0.0162
31.8	154.4	11.98	2249.	300.5	0.0000	0.0000	0.0181
32.0	154.4	11.95	2265.	300.5	0.0000	0.0000	0.0024
32.2	154.5	11.94	2295.	300.5	0.0000	0.0000	0.0035
32.4	154.3	11.93	2343.	300.5	0.0122	0.0000	0.0000
32.6	154.3	11.91	2389.	300.4	0.0063	0.0000	0.0000
32.8	154.3	11.92	2451.	300.4	0.0005	0.0000	0.0117
33.0	154.3	11.92	2511.	300.4	0.0000	0.0000	0.0670
33.2	154.4	11.90	2573.	300.4	0.0000	0.0207	0.0841
33.4	154.4	11.90	2639.	300.5	0.0000	0.0275	0.1189
33.6	154.5	11.87	2675.	300.5	0.0000	0.0520	0.1506
33.8	154.4	11.85	2717.	300.7	0.0000	0.0360	0.1913
34.0	154.5	11.84	2765.	300.6	0.0000	0.0437	0.2395
34.2	154.6	11.82	2834.	300.6	0.0000	0.0472	0.2597
34.4	154.5	11.82	2882.	300.5	0.0000	0.0979	0.2879
34.6	154.6	11.81	2948.	300.4	0.0225	0.1083	0.3178
34.8	154.4	11.79	2992.	300.5	0.0292	0.1302	0.3481
35.0	154.7	11.76	3074.	300.4	0.0154	0.1659	0.3785
35.2	154.8	11.73	3096.	300.3	0.0307	0.2112	0.3697
35.4	154.9	11.71	3079.	300.3	0.0239	0.2205	0.3837
35.6	155.0	11.79	3067.	300.4	0.0214	0.2236	0.3937
35.8	154.9	11.76	3062.	300.5	0.0194	0.2330	0.3986
36.0	155.2	11.75	3090.	300.5	0.0430	0.2434	0.3952
36.2	155.0	11.74	3100.	300.7	0.0264	0.2414	0.4101
36.4	155.0	11.72	3093.	300.7	0.0456	0.2526	0.4044
36.6	155.3	11.71	3074.	300.6	0.0353	0.2575	0.4116
36.8	155.1	11.73	3061.	300.6	0.0245	0.2635	0.3959
37.0	155.3	11.76	3092.	300.5	0.0342	0.2569	0.4114
37.2	155.0	11.78	3103.	300.5	0.0148	0.2609	0.4137
37.4	155.0	11.78	3104.	300.4	0.0562	0.2455	0.3910
37.6	155.0	11.78	3051.	300.5	0.0604	0.2331	0.3931
37.8	154.9	11.76	3052.	300.5	0.0558	0.2212	0.4059
38.0	155.1	11.76	3071.	300.6	0.0572	0.2256	0.4101
38.2	155.0	11.75	3073.	300.6	0.0351	0.2338	0.4144
38.4	155.1	11.75	3074.	300.6	0.0135	0.2481	0.3998
38.6	154.9	11.76	3089.	300.7	0.0053	0.2695	0.3952
38.8	154.9	11.76	3078.	300.6	0.0485	0.2631	0.3868
39.0	155.0	11.76	3062.	300.6	0.0312	0.2453	0.3982

B.2 DNB Measurements

Table B.2-1 Sample of steady-state DNB data in rod bundle in test series 4

No.	Test number	Pressure (kg/cm² a)	Mass flux (10^6 kg/m² hr)	Inlet temperature (℃)	DNB Power (MW)
1	04-3330	100.4	4.92	212.0	3.34
2	04-4330	125.1	4.93	233.1	3.05
3	04-3230	100.2	4.91	242.7	2.97
4	04-3240	100.5	7.93	242.0	3.87
5	04-3250	100.2	10.89	241.5	4.40
6	04-3760	100.5	14.09	262.4	4.36
7	04-3750	100.3	10.97	262.3	3.91
8	04-3730	99.9	4.95	263.8	2.85
9	04-4230	125.2	4.96	263.3	2.64
10	04-4250	125.3	10.92	261.8	4.21
11	04-4760	125.3	13.98	282.5	4.29
12	04-4750	125.3	10.95	282.6	3.61
13	04-4730	125.4	4.97	283.4	2.49
14	04-5230	150.1	4.91	281.8	2.36
15	04-5240	150.7	7.87	281.4	3.25
16	04-5250	150.4	10.87	281.5	4.11
17	04-5760	150.3	13.86	300.5	4.21
18	04-5750	150.5	10.84	300.7	3.50
19	04-5730	150.1	4.92	300.9	2.21
20	04-5130	150.1	4.89	321.5	1.87
21	04-5140	150.1	7.86	321.7	2.36
22	04-5150	150.0	10.87	321.3	2.78
23	04-5160	150.2	13.92	321.1	3.30
24	04-5170	150.4	16.78	321.0	3.77



Number: T/C # ○: Central ◇: Peripheral

Figure B.2-1 Sample of steady-state DNB detected location in rod bundle in test series 4

Fluid temperature was measured in each subchannel at 457 mm upper from the top end of the heated section in test assembly A1 without DNB condition in test series 1.

Table B.2-2 Sample of steady-state fluid temperature distribution in rod bundle in test series 1

No.	Test number	Pressure (kg/cm^2 a)	Mass flux (10^6 kg/m^2 hr)	Inlet temperature (℃)	Power (MW)
1	01-3233	100.0	4.62	166.1	0.98
2	01-3232	100.1	1.94	164.4	0.41
3	01-3239	100.1	1.24	165.3	0.25
4	01-3231	100.1	0.44	166.2	0.11
5	01-5343	150.3	5.03	165.3	1.25
6	01-5342	150.0	1.92	164.5	0.52
7	01-5242	149.9	1.92	185.1	0.41
8	01-5243	150.3	4.89	184.7	0.98
9	01-5143	150.2	4.66	202.9	0.70
10	01-5333	150.5	4.60	211.6	1.24

Fluid temperature distribution in rod bundle

RUN NO.1 Test number 01-3233 (Celsius)

	1	2	3	4	5	6
1	246.3	246.6	240.2	236.5	231.4	228.5
2	248.3	247.8	240.6	237.4	232.0	228.9
3	246.6	243.7	240.7	233.1	229.5	227.5
4	244.7	241.6	235.0	228.2	228.3	225.9
5	244.0	241.6	235.0	226.4	225.2	223.3
6	243.6	242.9	241.3	231.5	226.0	221.6

Table B.2-3 Sample of transient DNB data in rod bundle in test series 11T

Test number 11-0111 DNB detected time 92.1sec

Time (sec)	Pressure (kg/cm^2 a)	Mass flux (10^6 kg/m^2 hr)	Inlet temperature (℃)	Power (MW)
65	156.3	11.32	293.1	2.50
66	156.3	11.31	293.1	2.50
67	156.1	11.29	293.1	2.50
68	156.4	11.27	293.1	2.50
69	156.2	11.13	293.1	2.50
70	156.3	10.78	293.1	2.50
71	156.2	10.33	293.1	2.50
72	156.1	10.01	293.1	2.50
73	156.0	9.84	293.1	2.50
74	156.2	9.74	293.1	2.50
75	156.1	9.60	293.1	2.50
76	156.2	9.38	293.1	2.50
77	156.1	9.08	293.2	2.50
78	156.1	8.79	293.1	2.50
79	156.4	8.51	293.1	2.49
80	156.3	8.25	293.2	2.49
81	156.3	8.05	293.3	2.49
82	156.5	7.88	293.2	2.49
83	156.4	7.63	293.3	2.49
84	156.4	7.32	293.3	2.49
85	156.7	7.02	293.3	2.49
86	156.9	6.74	293.4	2.49
87	157.1	6.44	293.5	2.48
88	157.3	6.13	293.5	2.48
89	157.4	5.85	293.5	2.48
90	157.5	5.61	293.6	2.48
91	157.4	5.41	293.4	2.48
92	157.5	5.26	293.6	2.48
93	157.7	5.05	293.6	2.48
94	158.0	4.82	293.6	2.48